

Acetaldehyde

Acetaldehyde (IUPAC systematic name **ethanal**) is an organic chemical compound with the formula $\text{CH}_3\text{CH}=\text{O}$. It is a colorless liquid or gas, boiling near room temperature. It is one of the most important aldehydes, occurring widely in nature and being produced on a large scale in industry. Acetaldehyde occurs naturally in coffee, bread, and ripe fruit,^[11] and is produced by plants. It is also produced by the partial oxidation of ethanol by the liver enzyme alcohol dehydrogenase and is a contributing cause of hangover after alcohol consumption.^[12] Pathways of exposure include air, water, land, or groundwater, as well as drink and smoke.^[13] Consumption of disulfiram inhibits acetaldehyde dehydrogenase, the enzyme responsible for the metabolism of acetaldehyde, thereby causing it to build up in the body.

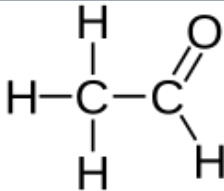
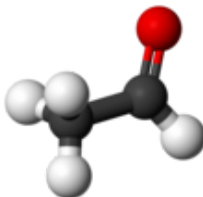
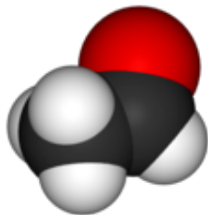
The International Agency for Research on Cancer (IARC) has listed acetaldehyde as a Group 1 carcinogen.^[14] Acetaldehyde is "one of the most frequently found air toxins with cancer risk greater than one in a million".^[15]

History

Acetaldehyde was first observed by the Swedish pharmacist/chemist Carl Wilhelm Scheele (1774);^[16] it was then investigated by the French chemists Antoine François, comte de Fourcroy and Louis Nicolas Vauquelin (1800),^[17] and the German chemists Johann Wolfgang Döbereiner (1821, 1822, 1832)^[18] and Justus von Liebig (1835).^{[19][20]}

In 1835, Liebig named it "aldehyde",^[21] and in the middle of the century the name was altered to "acetaldehyde".^[22]

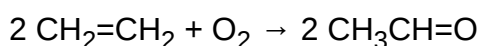
Acetaldehyde

	
	
Names	
Preferred IUPAC name	Acetaldehyde ^[3]
Systematic IUPAC name	Ethanal ^[3]
Other names	Acetic aldehyde Ethyl aldehyde ^[1] Acetylaldehyde ^[2]
Identifiers	
CAS Number	75-07-0 (https://commonchemistry.cas.org/detail?cas_rn=75-07-0) [✓]
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=O%3DCC) Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=CC%3DO)
ChEBI	CHEBI:15343 (https://www.ebi.ac.uk/chebi/searchId.do?chebiId=15343) [✓]
ChEMBL	ChEMBL170365 (https://www.ebi.ac.uk/chembl/explore/compound/ChEMBL170365) [✓]

Production

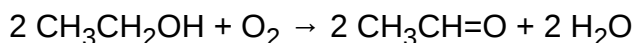
In 2013, global production was about 438,000 tonnes.^[23] Before 1962, ethanol and acetylene were the major sources of acetaldehyde. Since then, ethylene is the dominant feedstock.^[24]

The main method of production is the oxidation of ethene by the Wacker process, which involves oxidation of ethene using a homogeneous palladium/copper catalyst system:



In the 1970s, the world capacity of the Wacker-Hoechst direct oxidation process exceeded two million tonnes annually.

Smaller quantities can be prepared by the partial oxidation of ethanol in an exothermic reaction. This process typically is conducted over a silver catalyst at about 500–650 °C (950–1,200 °F).^[24]

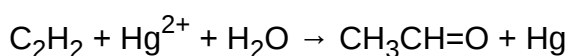


This method is one of the oldest routes for the industrial preparation of acetaldehyde.

Other methods

Hydration of acetylene

Prior to the Wacker process and the availability of cheap ethylene, acetaldehyde was produced by the hydration of acetylene.^[25] This reaction is catalyzed by mercury(II) salts:



The mechanism involves the intermediacy of vinyl alcohol, which tautomerizes to acetaldehyde. The reaction is conducted at 90–95 °C (194–203 °F), and the acetaldehyde formed is separated from water and mercury and cooled to 25–30 °C (77–86 °F). In the wet oxidation process, iron(III) sulfate is used to reoxidize

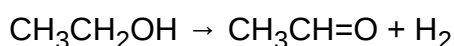
ChemSpider	172 (https://www.chemspider.com/Chemical-Structure.172.html) ✓
ECHA InfoCard	100.000.761 (https://echa.europa.eu/substance-information/-/substanceinfo/100.000.761)
EC Number	200-836-8
IUPHAR/BPS	6277 (http://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=6277)
KEGG	C00084 (https://www.kegg.jp/entry/C00084) ✓
PubChem CID	177 (https://pubchem.ncbi.nlm.nih.gov/compound/177)
RTECS number	AB1925000
UNII	GO1N1ZPR3B (https://precision.fda.gov/uniisearch/srs/unii/GO1N1ZPR3B) ✓
CompTox Dashboard (EPA)	DTXSID5039224 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID5039224)
InChI	InChI=1S/C2H4O/c1-2-3/h2H,1H3 ✓ Key: IKHGUXGNUITLKF-UHFFFAOYSA-N ✓
	InChI=1/C2H4O/c1-2-3/h2H,1H3 Key: IKHGUXGNUITLKF-UHFFFAOYAB
SMILES	O=CC CC=O
Properties	
Chemical formula	C ₂ H ₄ O
Molar mass	44.053 g·mol ^{−1}
Appearance	Colourless gas or liquid
Odor	Ethereal
Density	0.784 g·cm ^{−3} (20 °C) ^[4] 0.7904–0.7928 g·cm ^{−3} (10 °C) ^[4]

the mercury back to the mercury(II) salt. The resulting iron(II) sulfate is oxidized in a separate reactor with nitric acid.^[24]

The enzyme Acetylene hydratase discovered in the strictly anaerobic bacterium *Pelobacter acetylenicus* can catalyze an analogous reaction without involving any compounds of mercury.^[26] However, it has thus far not been brought to any large-scale or commercial use.

Dehydrogenation of ethanol

Traditionally, acetaldehyde was produced by the partial dehydrogenation of ethanol:



In this endothermic process, ethanol vapor is passed at 260–290 °C over a copper-based catalyst. The process was once attractive because of the value of the hydrogen coproduct,^[24] but in modern times is not economically viable.

Hydroformylation of methanol

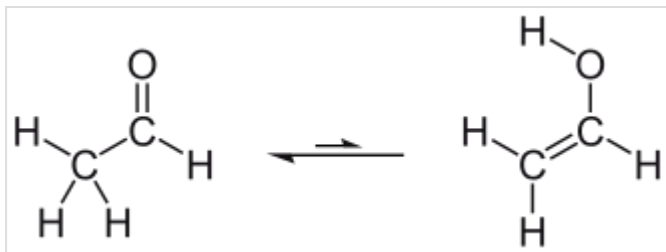
The hydroformylation of methanol with catalysts like cobalt, nickel, or iron salts also produces acetaldehyde, although this process is of no industrial importance. Similarly noncompetitive, acetaldehyde arises from synthesis gas with modest selectivity.^[24]

Melting point	−123.37 °C (−190.07 °F; 149.78 K)
Boiling point	20.2 °C (68.4 °F; 293.3 K)
Critical point (<i>T</i> , <i>P</i>)	466 K (193 °C), 5570 kPa
Solubility in water	miscible
Solubility	miscible with <u>ethanol</u> , <u>ether</u> , <u>benzene</u> , <u>toluene</u> , <u>xylene</u> , <u>turpentine</u> , <u>acetone</u> slightly soluble in <u>chloroform</u>
log <i>P</i>	−0.34
Vapor pressure	740 mmHg (20 °C) ^[5]
Acidity (<i>pK</i> _a)	13.57 (25 °C, H ₂ O) ^[6]
Magnetic susceptibility (<i>χ</i>)	−0.5153 ^{−6} cm ³ /g
Refractive index (<i>n</i> _D)	1.3316
Viscosity	0.21 mPa·s at 20 °C (0.253 mPa·s at 9.5 °C) ^[7]
Structure	
Molecular shape	trigonal planar (sp ²) at C ₁ tetrahedral (sp ³) at C ₂
Dipole moment	2.7 D
Thermochemistry ^[8]	
Heat capacity (<i>C</i>)	89 J·mol ^{−1} ·K ^{−1}
Std molar entropy (<i>S</i> [⊖] ₂₉₈)	160.2 J·mol ^{−1} ·K ^{−1}
Std enthalpy of formation (<i>Δ</i> _f <i>H</i> [⊖] ₂₉₈)	−192.2 kJ·mol ^{−1}
Gibbs free energy (<i>Δ</i> _f <i>G</i> [⊖])	−127.6 kJ·mol ^{−1}
Std enthalpy of combustion (<i>Δ</i> _c <i>H</i> [⊖] ₂₉₈)	−1167 kJ·mol
Enthalpy of fusion (<i>Δ</i> _f <i>H</i> [⊖] _{fus})	−2.310 kJ·mol
Enthalpy of vaporization (<i>Δ</i> _f <i>H</i> _{vap})	26.12 kJ·mol
Related compounds	
Related <u>aldehydes</u>	<u>Formaldehyde</u> <u>Propionaldehyde</u>
Related compounds	<u>Ethylene oxide</u>
Hazards	

Reactions

Tautomerization to vinyl alcohol

Occupational safety and health (OHS/OSH):	
Main hazards	potential occupational carcinogen ^[10]
GHS labelling:	
Pictograms	 ^[9]
Hazard statements	H224, H319, H335, H351 ^[9]
Precautionary statements	P210, P261, P281, P305+P351+P338 ^[9]
NFPA 704 (fire diamond)	
Flash point	−39.00 °C; −38.20 °F; 234.15 K
Autoignition temperature	175.00 °C; 347.00 °F; 448.15 K ^[5]
Explosive limits	4.0–60%
Lethal dose or concentration (LD, LC):	
LD ₅₀ (median dose)	1930 mg/kg (rat, oral)
LC ₅₀ (median concentration)	13,000 ppm (rat), 17,000 ppm (hamster), 20,000 ppm (rat) ^[10]
NIOSH (US health exposure limits):	
PEL (Permissible)	200 ppm (360 mg/m ³) ^[5]
IDLH (Immediate danger)	2000 ppm ^{[5][10]}
Safety data sheet (SDS)	HMDB (http://www.hmdb.ca/system/metabolites/msds/000/000/889/original/HMDB00990.pdf?1358893258)
Supplementary data page	
Acetaldehyde (data page)	
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa). ✓ verify (what is ✗ ?) Infobox references	



Tautomeric equilibrium between acetaldehyde and vinyl alcohol.

Like many other carbonyl compounds, acetaldehyde tautomerizes to give an enol (vinyl alcohol; IUPAC name: ethenol):



The equilibrium constant is 6×10^{-7} at room temperature, thus that the relative amount of the enol form in a sample of acetaldehyde is very small.^[27] At room temperature, acetaldehyde ($\text{CH}_3\text{CH}=\text{O}$) is more stable than vinyl alcohol ($\text{CH}_2=\text{CHOH}$) by 42.7 kJ/mol:^[28] Overall the keto-enol tautomerization occurs slowly but is catalyzed by acids.

Photo-induced keto-enol tautomerization is viable under atmospheric or stratospheric conditions. This photo-tautomerization is relevant to the Earth's atmosphere, because vinyl alcohol is thought to be a precursor to carboxylic acids in the atmosphere.^{[29][30]}

Addition and condensation reactions

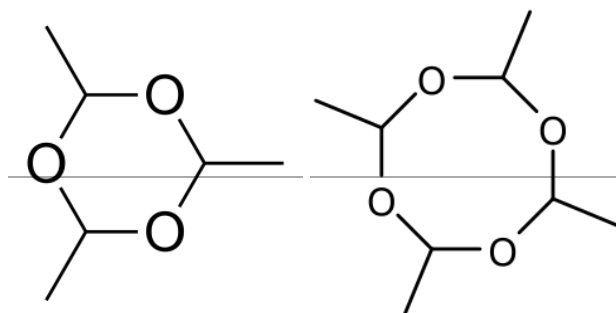
Acetaldehyde is a common electrophile in organic synthesis.^[31] In addition reactions acetaldehyde is prochiral. It is used primarily as a source of the " $\text{CH}_3\text{C}^+\text{H}(\text{OH})$ " synthon in aldol reactions and related condensation reactions.^[32] Grignard reagents and organolithium compounds react with MeCHO to give hydroxyethyl derivatives.^[33] In one of the more spectacular addition reactions, formaldehyde in the presence of calcium hydroxide adds to MeCHO to give pentaerythritol, $\text{C}(\text{CH}_2\text{OH})_4$ and formate.^[34]

In a Strecker reaction, acetaldehyde condenses with cyanide and ammonia to give, after hydrolysis, the amino acid alanine.^[35] Acetaldehyde can condense with amines to yield imines; for example, with cyclohexylamine to give N-ethylidenecyclohexylamine. These imines can be used to direct subsequent reactions like an aldol condensation.^[36]

It is also a building block in the synthesis of heterocyclic compounds. In one example, it converts, upon treatment with ammonia, to 5-ethyl-2-methylpyridine ("aldehyde-collidine").^[37]

Polymeric forms

Three molecules of acetaldehyde condense to form "paraldehyde", a cyclic trimer containing C-O single bonds. Similarly condensation of four molecules of acetaldehyde give the cyclic molecule metaldehyde. Paraldehyde can be produced in good yields, using a sulfuric acid catalyst. Metaldehyde is only obtained in a few percent yield and with cooling, often using HBr rather than H_2SO_4 as the catalyst. At -40°C (-40°F) in the presence of acid catalysts, polyacetaldehyde is produced.^[24] There are two stereoisomers of paraldehyde and four of metaldehyde.

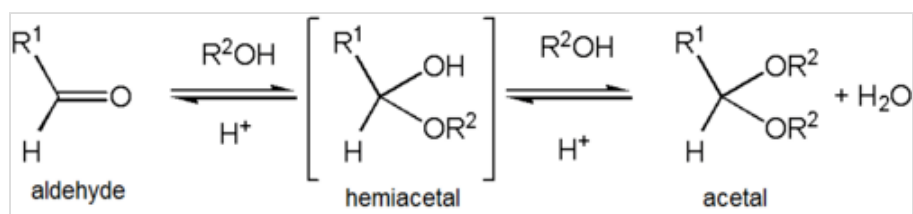


Cyclic oligomers of acetaldehyde (CH₃CHO)_n: paraldehyde (n = 3, left) and metaldehyde (n = 4, right)

The German chemist Valentin Hermann Weidenbusch (1821–1893) synthesized paraldehyde in 1848 by treating acetaldehyde with acid (either sulfuric or nitric acid) and cooling to 0 °C (32 °F). He found it quite remarkable that when paraldehyde was *heated* with a trace of the same acid, the reaction went the other way, recreating acetaldehyde.^[38]

Although vinyl alcohol is a tautomeric form of acetaldehyde (§ Tautomerization to vinyl alcohol), polyvinyl alcohol cannot be produced from acetaldehyde.

Acetal derivatives



Conversion of acetaldehyde to 1,1-diethoxyethane, R¹=CH₃ R²=CH₃CH₂

Acetaldehyde forms a stable acetal upon reaction with ethanol under conditions that favor dehydration. The product, CH₃CH(OCH₂CH₃)₂, is formally named 1,1-diethoxyethane but is commonly referred to as

"acetal".^[39] This can cause confusion as "acetal" is more commonly used to describe compounds with the functional groups RCH(OR')₂ or RR'C(OR'')₂ rather than referring to this specific compound — in fact, 1,1-diethoxyethane is also described as the diethyl acetal of acetaldehyde.

Precursor to vinylphosphonic acid

Acetaldehyde is a precursor to vinylphosphonic acid, which is used to make adhesives and ion conductive membranes. The synthesis sequence begins with a reaction with phosphorus trichloride:^[40]

1. $\text{PCl}_3 + \text{CH}_3\text{CHO} \rightarrow \text{CH}_3\text{CH}(\text{O}^-)\text{PCl}_3^+$
2. $\text{CH}_3\text{CH}(\text{O}^-)\text{PCl}_3^+ + 2 \text{CH}_3\text{CO}_2\text{H} \rightarrow \text{CH}_3\text{CH}(\text{Cl})\text{PO}(\text{OH})_2 + 2 \text{CH}_3\text{COCl}$
3. $\text{CH}_3\text{CH}(\text{Cl})\text{PO}(\text{OH})_2 \rightarrow \text{CH}_2=\text{CHPO}(\text{OH})_2 + \text{HCl}$

Biochemistry

In the liver, the enzyme, alcohol dehydrogenase oxidizes ethanol into acetaldehyde, which is then further oxidized into harmless acetic acid by acetaldehyde dehydrogenase. These two oxidation reactions are coupled with the reduction of NAD⁺ to NADH.^[41] In the brain, the enzyme catalase is primarily

responsible for oxidizing ethanol to acetaldehyde, and alcohol dehydrogenase plays a minor role.^[41] The last steps of alcoholic fermentation in bacteria, plants, and yeast involve the conversion of pyruvate into acetaldehyde and carbon dioxide by the enzyme pyruvate decarboxylase, followed by the conversion of acetaldehyde into ethanol. The latter reaction is again catalyzed by an alcohol dehydrogenase, now operating in the opposite direction.

Many East Asian people have an ALDH2 mutation which makes them significantly less efficient at oxidizing acetaldehyde. On consuming alcohol, their bodies tend to accumulate excessive amounts of acetaldehyde, causing the so-called alcohol flush reaction.^[42] They develop a characteristic flush on the face and body, along with "nausea, headache and general physical discomfort".^[43] Ingestion of the drug disulfiram, which inhibits ALDH2, leads to a similar reaction .^[44]

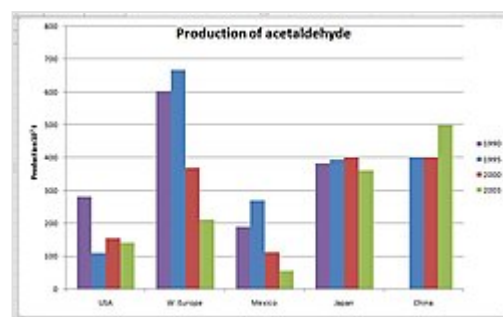
Uses

Traditionally, acetaldehyde was mainly used as a precursor to acetic acid. This application has declined because acetic acid is produced more efficiently from methanol by the Monsanto and Cativa processes. Acetaldehyde is an important precursor to pyridine derivatives, pentaerythritol, and crotonaldehyde. Urea and acetaldehyde combine to give a useful resin. Acetic anhydride reacts with acetaldehyde to give ethylidene diacetate, a precursor to vinyl acetate, which is used to produce polyvinyl acetate.^[24]

The global market for acetaldehyde is declining. Demand has been impacted by changes in the production of plasticizer alcohols, which has shifted because n-butyraldehyde is less often produced from acetaldehyde, instead being generated by hydroformylation of propylene. Likewise, acetic acid, once produced from acetaldehyde, is made predominantly by the lower-cost methanol carbonylation process.^[45] The impact on demand has led to increase in prices and thus slowdown in the market.

Consumption of acetaldehyde (10³ t) in 2003^[24]
 (* Included in others -glyoxal/glyoxalic acid, crotonaldehyde, lactic acid, *n*-butanol, 2-ethylhexanol)

Product	USA	Mexico	W. Europe	Japan	Total
Acetic Acid/Acetic anhydride	-	11	89	47	147
Acetate esters	35	8	54	224	321
Pentaerythritol	26	—	43	11	80
Pyridine and pyridine bases	73	—	10	*	83
Peracetic acid	23	—	—	*	23
1,3-Butylene glycol	14	—	—	*	14
Others	5	3	10	80	98
Total	176	22	206	362	766



Production of Acetaldehyde

China is the largest consumer of acetaldehyde in the world, accounting for almost half of global consumption in 2012. Major use has been the production of acetic acid. Other uses such as pyridines and pentaerythritol are expected to grow faster than acetic acid, but the volumes are not large enough to offset the decline in acetic acid. As a consequence, overall acetaldehyde consumption in China may grow slightly at 1.6% per year through 2018. Western Europe is the second-largest consumer of acetaldehyde worldwide, accounting for 20% of world consumption in 2012. As with China, the Western European acetaldehyde market is expected to increase only very slightly at 1% per year during 2012–2018. However, Japan could emerge as a potential consumer for acetaldehyde in the next five years due to newfound use in commercial production of butadiene. The supply of butadiene has been volatile in Japan and the rest of Asia. This should provide the much needed boost to the flat market, as of 2013.^[46]

Safety

Exposure limits

The threshold limit value is 25 ppm (STEL/ceiling value) and the MAK (Maximum Workplace Concentration) is 50 ppm. At 50 ppm acetaldehyde, no irritation or local tissue damage in the nasal mucosa is observed. When taken up by the organism, acetaldehyde is metabolized rapidly in the liver to acetic acid. Only a small proportion is exhaled unchanged. After intravenous injection, the half-life in the blood is approximately 90 seconds.^[24]

Dangers

Toxicity

Many serious cases of acute intoxication have been recorded.^[24] Acetaldehyde naturally breaks down in the human body.^{[13][47]}

Irritation

Acetaldehyde is an irritant of the skin, eyes, mucous membranes, throat, and respiratory tract. This occurs at concentrations as low as 1000 ppm. Symptoms of exposure to this compound include nausea, vomiting, and headache. These symptoms may not happen immediately. The perception threshold for acetaldehyde in air is in the range between 0.07 and 0.25 ppm.^[24] At such concentrations, the fruity odor of acetaldehyde is apparent. Conjunctival irritations have been observed after a 15-minute exposure to concentrations of 25 and 50 ppm, but transient conjunctivitis and irritation of the respiratory tract have been reported after exposure to 200 ppm acetaldehyde for 15 minutes.

Carcinogenicity

Acetaldehyde is carcinogenic in humans.^{[48][49]} In 1988 the International Agency for Research on Cancer stated, "There is *sufficient* evidence for the carcinogenicity of acetaldehyde (the major metabolite of ethanol) in experimental animals."^[50] In October 2009 the International Agency for Research on Cancer

updated the classification of acetaldehyde stating that acetaldehyde included in and generated endogenously from alcoholic beverages is a Group I human carcinogen.^[51] In addition, acetaldehyde is damaging to DNA^[52] and causes abnormal muscle development as it binds to proteins.^[53]

DNA crosslinks

Acetaldehyde induces DNA interstrand crosslinks, a form of DNA damage. These can be repaired by either of two replication-coupled DNA repair pathways.^[54] The first is referred to as the FA pathway, because it employs gene products defective in Fanconi's anemia patients. This repair pathway results in increased mutation frequency and altered mutational spectrum.^[54] The second repair pathway requires replication fork convergence, breakage of the acetaldehyde crosslink, translesion synthesis by a Y-family DNA polymerase and homologous recombination.^[54]

Aggravating factors

Alzheimer's disease

People with a genetic deficiency for the enzyme responsible for the conversion of acetaldehyde into acetic acid may have a greater risk of Alzheimer's disease. "These results indicate that the ALDH2 deficiency is a risk factor for LOAD [late-onset Alzheimer's disease] ..." ^[55]

Genetic conditions

A study of 818 heavy drinkers found that those exposed to more acetaldehyde than normal through a genetic variant of the gene encoding for ADH1C, ADH1C*1, are at greater risk of developing cancers of the upper gastrointestinal tract and liver.^[56]

Disulfiram

The drug disulfiram (Antabuse) inhibits acetaldehyde dehydrogenase, an enzyme that oxidizes the compound into acetic acid. Metabolism of ethanol forms acetaldehyde before acetaldehyde dehydrogenase forms acetic acid, but with the enzyme inhibited, acetaldehyde accumulates. If one consumes ethanol while taking disulfiram, the hangover effect of ethanol is felt more rapidly and intensely (disulfiram-alcohol reaction). As such, disulfiram is sometimes used as a deterrent for alcoholics wishing to stay sober.^[44]

Sources of exposure

Indoor air

Acetaldehyde is a potential contaminant in workplace, indoors, and ambient environments. Moreover, the majority of humans spend more than 90% of their time in indoor environments, increasing any exposure and the risk to human health.^[57]

In a study in France, the mean indoor concentration of acetaldehydes measured in 16 homes was approximately seven times higher than the outside acetaldehyde concentration. The living room had a mean of $18.1 \pm 17.5 \mu\text{g m}^{-3}$ and the bedroom was $18.2 \pm 16.9 \mu\text{g m}^{-3}$, whereas the outdoor air had a mean

concentration of $2.3 \pm 2.6 \mu\text{g m}^{-3}$.

It has been concluded that volatile organic compounds (VOC) such as benzene, formaldehyde, acetaldehyde, toluene, and xylenes have to be considered priority pollutants with respect to their health effects. It has been pointed that in renovated or completely new buildings, the VOCs concentration levels are often several orders of magnitude higher. The main sources of acetaldehydes in homes include building materials, laminate, PVC flooring, varnished wood flooring, and varnished cork/pine flooring (found in the varnish, not the wood). It is also found in plastics, oil-based and water-based paints, in composite wood ceilings, particle-board, plywood, treated pine wood, and laminated chipboard furniture.^[58]

Outdoor air

The use of acetaldehyde is widespread in different industries, and it may be released into waste water or the air during production, use, transportation and storage. Sources of acetaldehyde include fuel combustion emissions from stationary internal combustion engines and power plants that burn fossil fuels, wood, or trash, oil and gas extraction, refineries, cement kilns, lumber and wood mills and paper mills.^[59] Acetaldehyde is also present in automobile and diesel exhaust.^[60] As a result, acetaldehyde is "one of the most frequently found air toxics with cancer risk greater than one in a million".^[15]

Tobacco smoke

Natural tobacco polysaccharides, including cellulose, have been shown to be the primary precursors making acetaldehyde a significant constituent of tobacco smoke.^{[61][62]} It has been demonstrated to have a synergistic effect with nicotine in rodent studies of addiction.^{[63][64]} Acetaldehyde is also the most abundant carcinogen in tobacco smoke; it is dissolved into the saliva while smoking.

Cannabis smoke

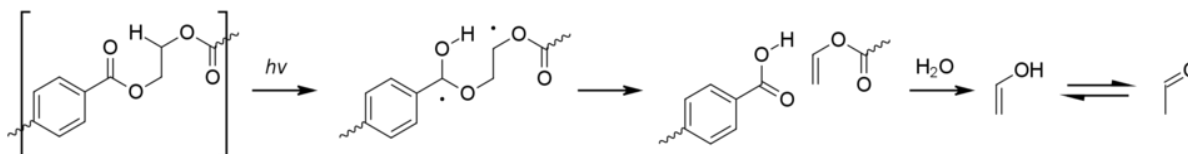
Acetaldehyde has been found in cannabis smoke. This finding emerged through the use of new chemical techniques that demonstrated the acetaldehyde present was causing DNA damage in laboratory settings.^[65]

Alcohol consumption

Many microbes produce acetaldehyde from ethanol, but they have a lower capacity to eliminate the acetaldehyde, which can lead to the accumulation of acetaldehyde in saliva, stomach acid, and intestinal contents. Fermented food and many alcoholic beverages can also contain significant amounts of acetaldehyde. Acetaldehyde, derived from mucosal or microbial oxidation of ethanol, tobacco smoke, and diet, appears to act as a cumulative carcinogen in the upper digestive tract of humans.^[66] According to European Commission's Scientific Committee on Consumer Safety's (SCCS) "Opinion on Acetaldehyde" (2012) the cosmetic products special risk limit is 5 mg/L and acetaldehyde should not be used in mouth-washing products.^[67]

Plastics

Acetaldehyde can be produced by the photo-oxidation of polyethylene terephthalate (PET), via a Type II Norrish reaction.^[68]



Although the levels produced by this process are minute acetaldehyde has an exceedingly low taste/odor threshold of around 20–40 ppb and can cause an off-taste in bottled water.^[69] The level at which an average consumer could detect acetaldehyde is still considerably lower than any toxicity.^[70]

Candida overgrowth

The yeast *Candida albicans* in patients with potentially carcinogenic oral diseases has been shown to produce acetaldehyde in quantities sufficient to cause problems.^[71]

See also

- Alcohol dehydrogenase
- Disulfiram-like drug
- Formaldehyde
- Paraldehyde
- Wine fault

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 - Dabit, a pharmacist in Nantes, France, performed a series of experiments and concluded that acetaldehyde was formed when hydrogen in ethanol combined with oxygen in sulfuric acid to form water: Dabit (1800). "Extrait du mémoire du cit. Dabit sur l'éther" (<https://books.google.com/books?id=PDZQAAAACAAJ&pg=PA289>) [Extract of the memoir by citizen Dabit on ether]. *Annales de Chimie*. **34**: 289–305.
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